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C4 Architecture Diagram Pack

Common IIP Context Used

The Insurance Intelligence Platform (IIP) is an Azure-native, human-reviewed insurance operations platform for African insurers and related insurance organizations. It acts as an intelligent operational layer across staff, customers, core insurance systems, documents, channels, knowledge sources, and AI agents. The current MVP direction is Motor Claims & Document Intelligence for a mid-to-large insurer pilot.

Confirmed implementation direction:

- Frontend: Next.js, React, TypeScript, Tailwind CSS.
- Backend: .NET 8 / ASP.NET Core, EF Core, Azure SQL.
- Azure services: Azure SQL, Blob Storage, Key Vault, Azure AI Search, Azure OpenAI, Azure AI Document Intelligence, Service Bus or Event Grid, Functions or Durable Functions, API Management, Application Insights and Log Analytics.
- Identity: Microsoft Entra ID / MSAL / OAuth 2.0.
- Primary internal modules: Command Center, Claims, Documents, Channels, Customers, Sales & Leads, Knowledge Base, Approvals, Compliance & Audit, Reports & Analytics, Integrations, Admin Settings, Support & Health.
- Customer entry points: WhatsApp-first, with web chat, SMS, email, broker, call center, Teams, and website forms as planned or future channels.
- Governance rule: AI prepares insurance work; humans make consequential decisions.

Non-negotiable AI boundary:

- AI may summarize, classify, extract, flag, draft, route, and recommend.
- AI must not approve or reject claims, bind policies, set final premiums, accuse customers of fraud, release payments, promise settlement, or make final regulated decisions.

Purpose

This pack provides Mermaid-based architecture diagrams for IIP. These diagrams are intended for technical alignment, implementation planning, stakeholder explanation, and future conversion into visual diagrams.

Diagram 1: C4 Context

```
flowchart TB
  Staff[Insurance Staff: Claims, Underwriting, Compliance, Customer Service]
  Customer[Customers, Brokers, Agents]
  IIP[Insurance Intelligence Platform]
  Core[Core Insurance / Policy / Claims Systems]
  M365[Microsoft 365 / Graph / Outlook / Teams]
  WhatsApp[WhatsApp Business Provider]
  Azure[Azure Platform Services]
  Regulators[Regulators / Auditors]

  Customer --> WhatsApp
  WhatsApp --> IIP
  Staff --> IIP
  IIP <--> Core
  IIP <--> M365
  IIP --> Azure
  IIP --> Regulators
```

Diagram 2: Container View

```
flowchart LR
  Web[Next.js Staff App]
  API[.NET 8 API]
  Worker[Functions / Durable Workers]
  SQL[(Azure SQL)]
  Blob[(Blob Storage)]
  Search[(Azure AI Search)]
  OpenAI[Azure OpenAI]
  DocAI[Azure AI Document Intelligence]
  Bus[Service Bus / Event Grid]
  KeyVault[Key Vault]
  AppInsights[Application Insights / Log Analytics]

  Web --> API
  API --> SQL
  API --> Blob
  API --> Search
  API --> OpenAI
  API --> DocAI
  API --> Bus
  Worker --> Bus
  Worker --> SQL
  Worker --> Blob
  Worker --> Search
  API --> KeyVault
  Worker --> KeyVault
  API --> AppInsights
  Worker --> AppInsights
```

Diagram 3: Component View - Backend API

```
flowchart TB
  Controllers[API Controllers]
  Auth[Authentication and Tenant Context Middleware]
  Claims[Claims Service]
  Docs[Document Service]
  Channels[Channel Service]
  AI[AI Orchestration Service]
  Approvals[Approval Service]
  Audit[Audit Service]
  Integrations[Integration Adapter Layer]
  Repo[EF Core Repositories]

  Controllers --> Auth
  Controllers --> Claims
  Controllers --> Docs
  Controllers --> Channels
  Controllers --> AI
  Controllers --> Approvals
  Claims --> Audit
  Docs --> Audit
  AI --> Audit
  Approvals --> Audit
  Claims --> Repo
  Docs --> Repo
  Channels --> Repo
  Integrations --> Repo
```

Diagram 4: Deployment View - Managed Application Path

```
flowchart TB
  Pub[Otic Publisher Tenant]
  Cust[Customer Azure Tenant]
  RG[Managed Resource Group]
  App[Container Apps / App Service]
  DB[(Customer Azure SQL)]
  Storage[(Customer Blob Storage)]
  KV[Customer Key Vault]
  Monitor[Customer Monitoring]
  Support[Controlled Publisher Support Access]

  Pub -->|Managed App Package| Cust
  Cust --> RG
  RG --> App
  RG --> DB
  RG --> Storage
  RG --> KV
  RG --> Monitor
  Support -.approved access.-> RG
```

Diagram 5: Deployment View - SaaS Path

```
flowchart TB
  TenantA[Insurer Tenant A]
  TenantB[Insurer Tenant B]
  Web[Shared SaaS Frontend]
  API[Shared SaaS API]
  Data[(Tenant-Isolated Data Stores)]
  Search[(Tenant-Isolated Search Indexes or Filters)]
  Storage[(Tenant-Isolated Storage Containers)]
  Ops[Otic Operations]

  TenantA --> Web
  TenantB --> Web
  Web --> API
  API --> Data
  API --> Search
  API --> Storage
  Ops --> API
```

Diagram 6: AI/RAG Flow

```
flowchart LR
  User[Staff User] --> Prompt[Question / Task]
  Prompt --> Guardrails[Policy and Role Guardrails]
  Guardrails --> Retrieval[Azure AI Search Retrieval]
  Retrieval --> Sources[Approved Knowledge and Claim Evidence]
  Sources --> Model[Azure OpenAI]
  Model --> Output[Grounded Draft / Summary / Recommendation]
  Output --> Review[Human Review]
  Review --> Audit[Audit Log]
```

Diagram 7: Document Intelligence Flow

```
flowchart LR
  Upload[Document Upload] --> Blob[Blob Storage]
  Upload --> Metadata[Document Metadata]
  Metadata --> Classify[Classify Document Type]
  Classify --> Extract[Extract Fields]
  Extract --> Validate[Validate and Cross-check]
  Validate --> Review[Human Review]
  Review --> Claim[Claim Case Evidence]
  Review --> Audit[Audit Events]
```

Diagram 8: Integration Flow

```
flowchart LR
  Channel[WhatsApp / Email / Web] --> Webhook[Webhook Receiver]
  Webhook --> Idempotency[Idempotency and Tenant Resolution]
  Idempotency --> Queue[Service Bus / Event Grid]
  Queue --> Worker[Integration Worker]
  Worker --> Case[Claim / Customer / Lead Services]
  Case --> Audit[Audit Log]
```

Diagram Usage Notes

Diagram	Use Case
Context	Executive and client technical orientation.
Container	Engineering and Azure architecture planning.
Component	Backend implementation planning.
Managed Application	Enterprise customer tenant deployment discussion.
SaaS	Otic-hosted option and multi-tenant design discussion.
AI/RAG	AI safety, grounding, and governance review.
Document Intelligence	OCR/extraction workflow implementation.
Integration Flow	Channel and webhook implementation.

Implementation Implications

- Tenant context must be resolved before any data access.

- AI services must be wrapped by an orchestration layer; frontend must not call model services directly.
- Document storage, extraction, review, and audit should be separate but linked processes.
- Integration webhooks require idempotency, retries, correlation IDs, and audit logging.
- Managed Application and SaaS paths share core application logic but differ in tenancy, provisioning, support, and deployment boundaries.

Open Questions

ID	Open Question	Owner	Impact if Unresolved
OQ-01	Which insurer will be the first signed pilot customer and what exact internal processes will be included?	Product / Commercial	Final configuration, training, and pilot timelines may change.
OQ-02	Which claims/core system will be integrated first, and will MVP require read-only lookup, export, or direct write-back?	Product / Engineering / Client IT	Impacts integration complexity and deployment schedule.
OQ-03	Which Azure deployment path is selected first: customer-tenant Managed Application or Otic-hosted SaaS pilot?	Leadership / Engineering	Impacts tenancy, support access, IaC, billing, and security review.
OQ-04	Which WhatsApp Business provider will be used for the first pilot?	Product / Integrations	Impacts webhook schemas, retry behavior, and channel onboarding.
OQ-05	What client-specific SLAs, roles, branches, claim categories, and document checklists must be configured?	Client Operations	Impacts workflow rules and acceptance testing.
OQ-06	What final regulatory, legal, and data residency requirements must be signed off for the first pilot?	Legal / Compliance	Impacts production readiness and contractual commitments.